

Network Traffic Analysis using Wireshark

Project Report

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Project Type: Cybersecurity / Network Traffic Analysis

Tools Used: Wireshark, Nmap, Windows Command Prompt

1. Introduction

Network traffic analysis is an important aspect of cybersecurity and network monitoring. By capturing and analyzing packets transmitted over a network, analysts can understand communication behavior, identify protocols in use, detect suspicious activity, and investigate security-related events.

This project focuses on analyzing live network traffic using Wireshark. Various protocols such as TCP, DNS, HTTP, HTTPS/TLS, and ICMP were examined to understand how systems communicate over networks. Additionally, suspicious traffic patterns such as repeated ICMP requests and SYN scan activity were simulated and analyzed.

2. Objective

The objectives of this project were:

- Capture and analyze live network traffic
- Study protocol behavior and packet structure
- Understand TCP connection establishment
- Analyze DNS query and response behavior
- Compare HTTP and HTTPS communication
- Observe encrypted TLS traffic
- Detect repeated ICMP traffic patterns
- Simulate and analyze SYN scan activity using Nmap

3. Tools Used

Tool	Purpose
Wireshark	Packet capture and protocol analysis
Nmap	Port scanning and SYN scan generation
Windows Command Prompt	ICMP traffic generation

4. TCP 3-Way Handshake Analysis

TCP uses a three-step handshake process to establish reliable communication between a client and server.

The process includes:

1. SYN – Client initiates connection request
2. SYN ACK – Server acknowledges the request
3. ACK – Client confirms the connection

This handshake establishes a reliable communication channel before data transfer begins.

Observations

- TCP packets were filtered using the tcp filter in Wireshark
- SYN, SYN ACK, and ACK packets were identified
- HTTPS communication over port 443 was observed
- Packet details such as sequence numbers, acknowledgment numbers, and ports were analyzed

Screenshot

tcp						
No.	Time	Source	Destination	Protocol	Length	Info
5	1.386245	172.20.10.3	52.72.119.211	TCP	55	54043 → 5228 [ACK] Seq=1 Ack=1 Win=253 Len=1
6	1.579646	2401:4900:96ad:cb36...	2404:6800:4003:c06...	TCP	75	5228 → 54043 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
8	1.668781	2404:6800:4003:c06...	2401:4900:96ad:cb36...	TCP	86	5228 → 54043 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
9	1.678199	100.31.234.221	172.20.10.3	TCP	66	443 → 51634 [ACK] Seq=1 Ack=2 Win=425 Len=0 SLE=1 SRE=2
10	1.908973	52.72.119.211	172.20.10.3	TCP	66	443 → 50887 [ACK] Seq=1 Ack=2 Win=136 Len=0 SLE=1 SRE=2
...	...	...	...	...	...	...
46	9.747696	18.161.232.19	172.20.10.3	TCP	66	443 → 60779 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1300 SACK_PERM WS=512
47	9.747698	172.20.10.3	18.161.232.19	TCP	54	60779 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
...	...	...	...	...	...	...
59	9.834618	172.20.10.3	18.161.232.19	TCP	201	Client Hello (SNI=ezx15jj5maj1.vxn.cloudfront.net)
60	9.877934	18.161.232.19	172.20.10.3	TCP	645	Server Hello, Change Cipher Spec, Encrypted Handshake Message
61	9.878761	18.161.232.19	172.20.10.3	TCP	162	Change Cipher Spec, Encrypted Handshake Message, Application Data
62	9.948337	172.20.10.3	18.161.232.19	TCP	54	443 → 60779 [ACK] Seq=148 Ack=1055 Win=68096 Len=0
63	10.07144	18.161.232.19	172.20.10.3	TLSv1.2	555	Application Data
64	10.12287	172.20.10.3	18.161.232.19	TCP	54	60779 → 443 [ACK] Seq=1055 Ack=649 Win=64768 Len=0
▶ Frame 47: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{DBD7C8F7-F4FB-454A-83DD-D14C29331}						
▶ Ethernet II, Src: Intel_b5:35:29 (04:ec:d8:b5:35:29), Dst: fa:42:88:35:27:64 (fa:42:88:35:27:64)						
▶ Internet Protocol Version 6, Src: 2401:4900:96ad:cb36:b4d9:cd6:cb12:3681, Dst: 2606:4700:83b1:db87:9afd:2a:ccf5:ac50						
▶ Transmission Control Protocol, Src Port: 60779, Dst Port: 443, Seq: 1, Ack: 1, Len: 0						
0000	fa 42 88 35 27 64 04 ec d8 b5 35 29 86 dd 60 05	..B.5'.d....3)....'				
0010	a5 10 00 14 06 ff f2 02 :0 40 40 06 3b 6a b4 d9	\$.@ @. j..				
0020	cd 6c cb 12 36 81 26 06 47 00 83 b1 db 87 9a fd	..l...6.&..G.....				
0030	2a cc cf 5a ac 50 ef 3b 01 bb dc d5 f3 43 df a1	*...Z.P.;;.....C..				
0040	81 21 50 10 00 ff 70 21 00 00	..!P...p!...				

5. TCP Stream Analysis

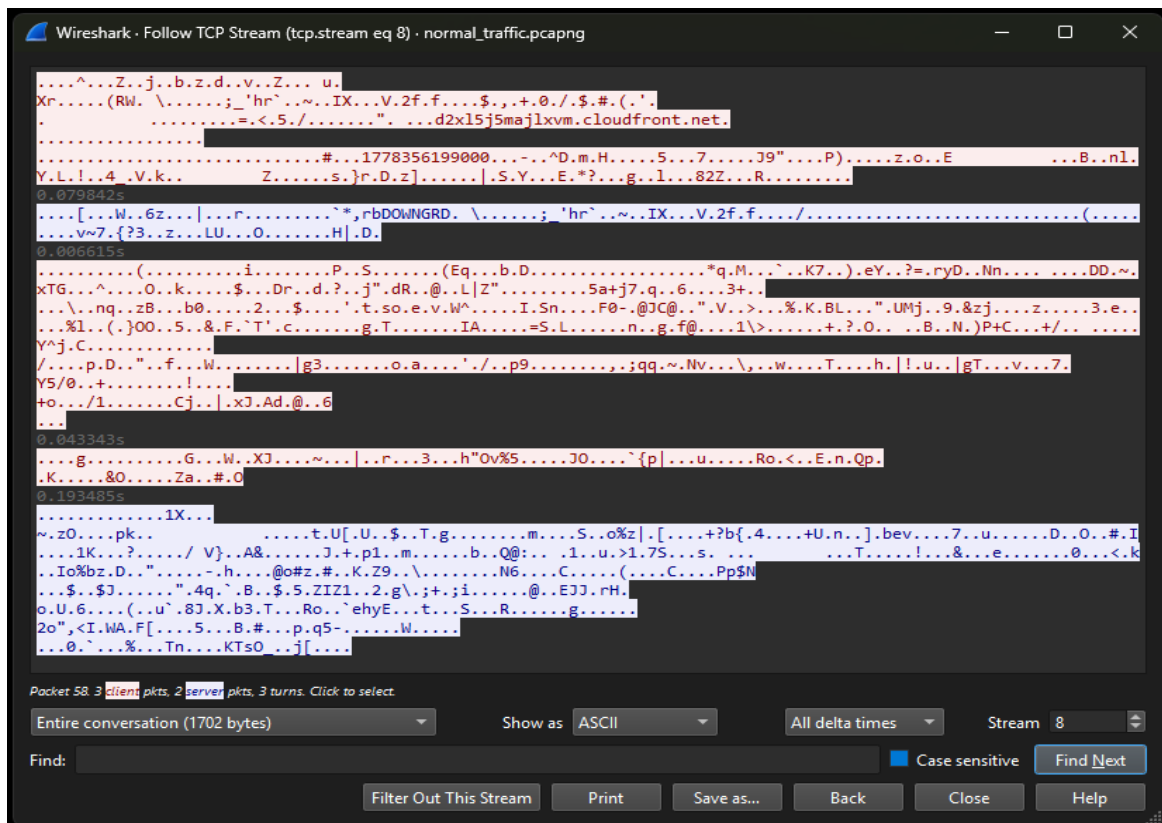
Wireshark's "Follow TCP Stream" feature was used to reconstruct communication between the client and server.

The captured traffic showed encrypted HTTPS/TLS communication. Since the traffic was encrypted, most payload data appeared unreadable in plaintext format.

Observations

- Communication sessions between systems were reconstructed
- Encrypted payload data was observed
- Domain references such as CloudFront were visible
- Secure communication behavior was analyzed

Screenshot



6. DNS Query Analysis

DNS (Domain Name System) is responsible for translating domain names into IP addresses required for communication.

Wireshark was used to observe DNS query and response packets generated during web browsing.

Observations

- Standard DNS queries and responses were captured
- Domains such as cloudfront.net, googleapis.com, and googleusercontent.com were resolved
- Returned IPv4 and IPv6 addresses were analyzed
- DNS response behavior was examined

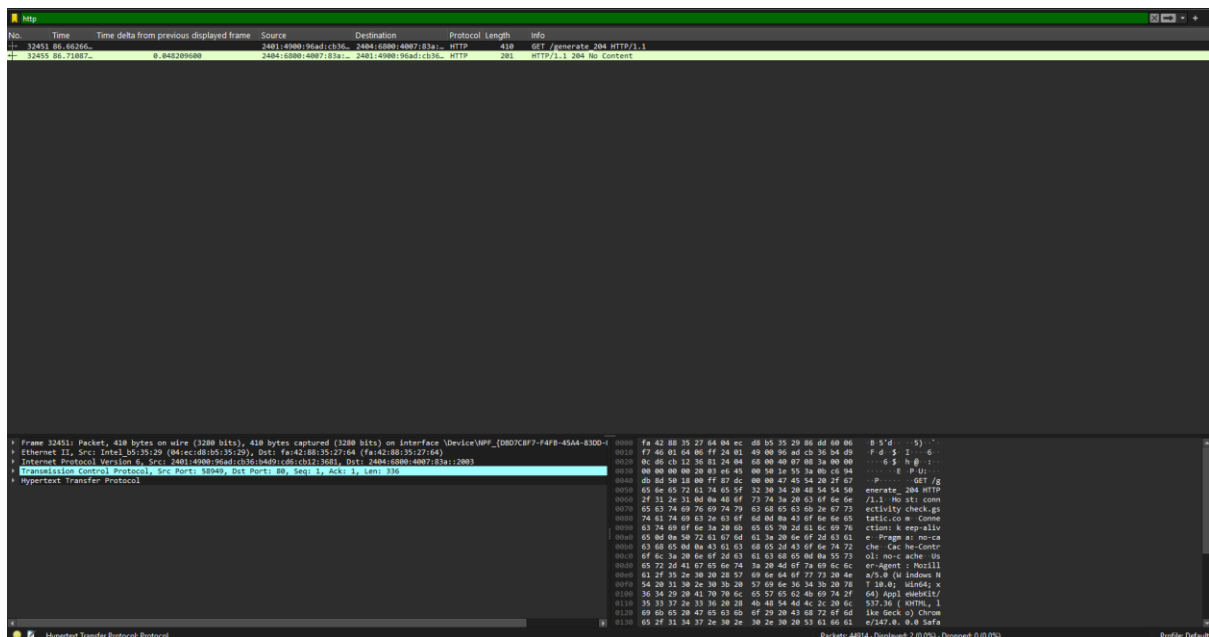
No.	Time	Time delta from previous displayed frame	Source	Destination	Protocol	Length	Info
49	0.640204		f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	189	Standard query 0x6d4f a.d215155jswaii.cloudflare.net
50	0.645355	0.000525030	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	189	Standard query 0x6d4e AAAA d215155jswaii.cloudflare.net
51	0.687515	0.055236000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	194	Standard query response 0x6d4e AAAA d215155jswaii.cloudflare.net ns-11380.madison.ny.us
52	0.723113700	0.035598700	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	193	Standard query response 0x6d4e AAAA d215155jswaii.cloudflare.net ns-11381.229.19.181.161.232.20.9.18.161.232.20.9.18.161.232.20.9
53	1.300177014	0.000131120	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	249	Standard query response 0x6d4e HTTPS 607.org HTTPS
54	1.300177014	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	113	Standard query response 0x6d4f a.d07.org a 172.67.155.178 A 188.128.28.225
55	1.300177014	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	112	Standard query response 0x6d4f AAAA d215155jswaii.cloudflare.net ns-11380.madison.ny.us
56	1.37793822	0.020883100	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	189	Standard query 0x6ffca AAAA agads-pa.clients.google.com AAAA 2484-6880-4807:8161:200a
57	1.37793822	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	194	Standard query response 0x6ffca AAAA agads-pa.clients.google.com AAAA 2484-6880-4807:8161:200a
58	1.80718267	0.050551500	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	173	Standard query response 0x6ffca d2151552mks.cloudflare.net a 108.159.51.14 A 108.159.51.14 A 108.159.51.14 A 108.159.51.137
59	2.12017472	0.000002400	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	194	Standard query response 0x6ff9 AAAA d2151552mks.cloudflare.net SOA ns-11313.madison.ny.us
60	2.12017472	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	172	Standard query response 0x6ff9 AAAA d2151552mks.cloudflare.net ns-11313.madison.ny.us
61	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	194	Standard query response 0x6ff9 AAAA agads-pa.clients.google.com AAAA 2484-6880-4807:8161:200a
62	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	164	Standard query response 0x6ff9 A 193.googleusercontent.com CNAME googlehosted-1.googleusercontent.com A 142.251.229.97
63	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	194	Standard query response 0x6ff9 AAAA agads-pa.clients.google.com AAAA 2484-6880-4807:8161:200a
64	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	194	Standard query response 0x6ff9 HTTPS agads-pa.clients.google.com SOA ns-193.google.com
65	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	205	Standard query response 0x6ff9 HTTPS 193.googleusercontent.com CNAME googlehosted-1.googleusercontent.com SOA ns-193.google.com
66	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	172	Standard query response 0x6ff9 AAAA agads-pa.clients.google.com A 142.251.27.138
67	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	140	Standard query response 0x6ff9 AAAA encrypted-thb1.gttatic.com AAAA 2484-6880-4807:8151:200a
68	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	177	Standard query response 0x6ff9 HTTPS encrypted-thb1.gttatic.com SOA ns-193.google.com
69	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	136	Standard query response 0x6ff9 A encrypted-thb1.gttatic.com A 142.258.163.78
70	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	159	Standard query response 0x6ff9 HTTPS play.google.com SOA ns-193.google.com
71	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	137	Standard query response 0x6ff9 AAAA 193.googleusercontent.com CNAME googlehosted-1.googleusercontent.com AAAA 2484-6880-4807:8161:200a
72	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	155	Standard query response 0x6ff9 A play.google.com A 172.217.214.214
73	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	193	Standard query response 0x6ff9 A optimizationipide-pa.googleapis.com A 216.239.32.223 A 216.239.38.223 A 216.239.38.223 A 216.239.34.223
74	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	193	Standard query response 0x6ff9 AAAA optimizationipide-pa.googleapis.com SOA ns-193.google.com
75	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	241	Standard query response 0x6ff9 AAAA optimizationipide-pa.googleapis.com AAAA 2081-4880-4802:161:223 AAAA 2081-4880-4802:161:223 AAAA 2081-4880-4802:161:223 AAAA 2081-4880-4802:161:223
76	3.081441400	0.000000000	f800::f8a2::bfff::fe	f800::f8a2::bfff::fe	DNS	205	Standard query response 0x6ff9 HTTPS 193.googleusercontent.com CNAME googlehosted-1.googleusercontent.com SOA ns-193.google.com
77	3.081441400	0.000000000	f800::f8a2::bfff::				

HTTP traffic was analyzed to observe plaintext communication behavior.

Observations

- HTTP GET requests were visible
- Packet headers and metadata could be inspected
- Plaintext communication was identified
- Limited HTTP traffic was observed because most modern websites use HTTPS

Screenshot



8. HTTPS / TLS Traffic Analysis

HTTPS traffic uses TLS encryption to secure communication between systems.

Wireshark captured TLSv1.2 and TLSv1.3 traffic generated during secure browsing activity.

Observations

- TLSv1.2 and TLSv1.3 packets were observed
- Change Cipher Spec messages were captured
- Encrypted Application Data packets were identified
- HTTPS communication over port 443 was analyzed
- Payload encryption prevented direct plaintext inspection

[illegible]

ICMP (Internet Control Message Protocol) is commonly used for network diagnostics and connectivity testing.

Observations

- Echo Request packets were sent continuously
- Echo Reply packets were received from the destination server
- Repeated ICMP traffic patterns were observed
- Sequence numbers and TTL values were analyzed

Repeated ICMP traffic may also indicate:

- Network scanning
- Reconnaissance activity
- ICMP flooding attempts

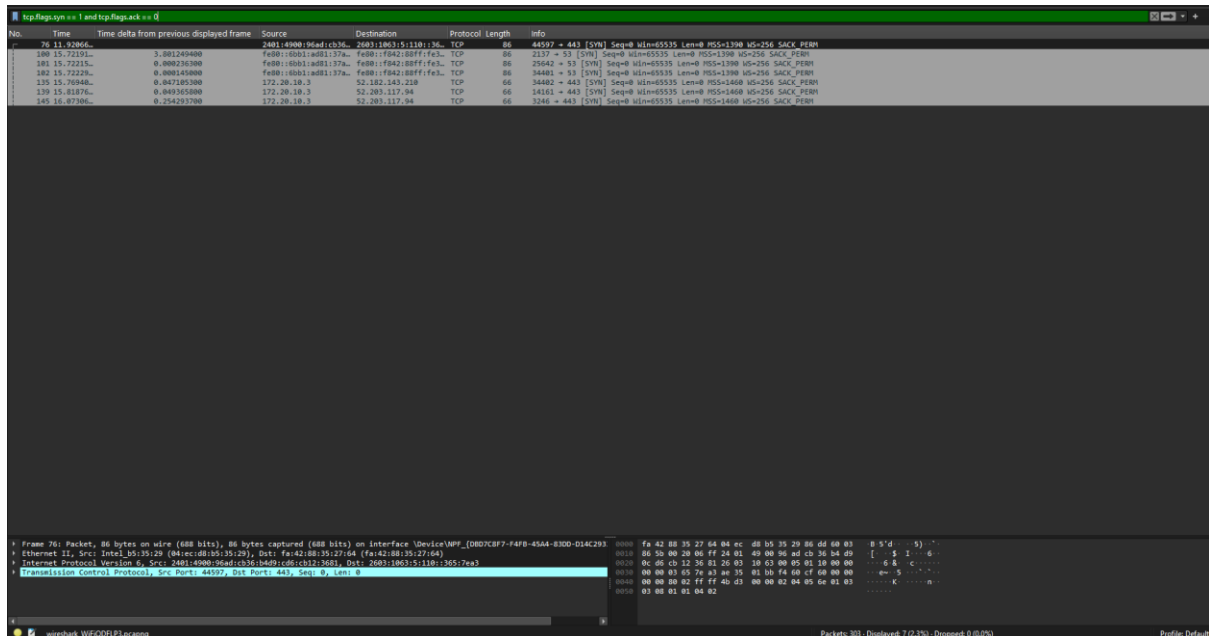
No.	Time	Time delta from previous displayed frame	Source	Destination	Protocol	Length	Info
115	9.421171		172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=1256, ttl=128 (reply in 270)
116	9.421171	0.450670100	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=1256, ttl=128 (reply in 145)
254	10.433232	0.951533900	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=2512, ttl=128 (reply in 121)
271	10.408662	0.844295900	142.251.222.206	172.28.18.3	ICMP	74	Echo (ping) reply id=60001, seq=2512, ttl=116 (request in 254)
272	10.450451	0.290451100	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=3768, ttl=128 (reply in 293)
293	11.524086	0.805794400	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) reply id=60001, seq=3768, ttl=116 (request in 292)
295	12.446623	0.922764200	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=4184, ttl=128 (reply in 296)
296	12.515371	0.866861500	142.251.222.206	172.28.18.3	ICMP	74	Echo (ping) reply id=60001, seq=4184, ttl=116 (request in 295)
299	13.420782	0.940488000	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=51280, ttl=128 (reply in 380)
300	13.526477	0.864793100	142.251.222.206	172.28.18.3	ICMP	74	Echo (ping) reply id=60001, seq=51280, ttl=116 (request in 299)
329	14.462558	0.939027000	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=61536, ttl=128 (reply in 344)
340	14.526190	0.062292100	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=61536, ttl=128 (request in 329)
359	15.477444	0.948594400	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=71792, ttl=128 (reply in 397)
397	15.561119	0.887953100	142.251.222.206	172.28.18.3	ICMP	74	Echo (ping) reply id=60001, seq=71792, ttl=116 (request in 392)
405	16.925770	0.820577000	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=82048, ttl=128 (reply in 481)
406	16.551133	0.863409200	142.251.222.206	172.28.18.3	ICMP	74	Echo (ping) reply id=60001, seq=82048, ttl=116 (request in 485)
506	17.585656	0.952477900	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=92304, ttl=128 (reply in 508)
508	17.593118	0.847561100	142.251.222.206	172.28.18.3	ICMP	74	Echo (ping) reply id=60001, seq=92304, ttl=116 (request in 508)
540	18.510744	0.959595900	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=102560, ttl=128 (reply in 540)
549	18.577651	0.861764600	142.251.222.206	172.28.18.3	ICMP	74	Echo (ping) reply id=60001, seq=102560, ttl=116 (request in 548)
552	19.521233	0.948729000	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=112816, ttl=128 (reply in 553)
553	19.596189	0.074956100	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=112816, ttl=128 (reply in 552)
559	19.540105	0.936969500	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=127807, ttl=128 (reply in 560)
560	20.643448	0.183273000	142.251.222.206	172.28.18.3	ICMP	74	Echo (ping) reply id=60001, seq=127807, ttl=116 (request in 559)
565	21.551346	0.951346000	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=137328, ttl=128 (reply in 565)
566	21.551133	0.951599000	142.251.222.206	172.28.18.3	ICMP	74	Echo (ping) reply id=60001, seq=137328, ttl=116 (request in 565)
570	22.561846	0.951247000	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=147354, ttl=128 (reply in 570)
570	22.644828	0.882415400	142.251.222.206	172.28.18.3	ICMP	74	Echo (ping) reply id=60001, seq=147354, ttl=116 (request in 569)
577	23.572498	0.927688800	172.28.18.3	142.251.222.206	ICMP	74	Echo (ping) request id=60001, seq=157840, ttl=128 (reply in 583)
583	23.613137	0.040663300	142.251.222.206	172.28.18.3	ICMP	74	Echo (ping) reply id=60001, seq=157840, ttl=116 (request in 577)
594	24.579232	0.960408000	172.28.18.3				

[illegible]

- Penetration testing

Excessive SYN activity may also indicate suspicious scanning behavior.

Screenshot



11. Key Learnings

Through this project, the following concepts were understood:

- Packet capture and traffic monitoring
- TCP communication workflow
- DNS resolution process
- Difference between HTTP and HTTPS
- Importance of TLS encryption
- ICMP diagnostic behavior
- Basic reconnaissance and scanning techniques
- Practical usage of Wireshark filters
- Security monitoring concepts relevant to SOC environments

12. Conclusion

This project provided practical exposure to network traffic analysis and protocol monitoring using Wireshark.

Multiple network protocols including TCP, DNS, HTTP, HTTPS/TLS, and ICMP were analyzed to understand how systems communicate across networks. In addition, suspicious traffic patterns such as repeated ICMP requests and SYN scan activity were generated and investigated.

The project strengthened foundational cybersecurity concepts related to network monitoring, packet inspection, protocol analysis, and basic threat detection techniques. The skills gained through this project are directly relevant to SOC analysis, network security, and cybersecurity operations.